

KAMYSHIN, Russian Federation

WMO: 343630

Lat: 50.083N

Lon: 45.367E

Elev: 389

StdP: 14.49

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.3	-4.6	-15.6	2.4	-8.4	-10.7	3.1	-3.9	31.4	19.6	27.4	23.8	9.9	30	0.613

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	95.0	66.9	91.6	66.4	88.1	65.4	70.1	86.4	68.7	84.6	67.5	83.0	10.0	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.0	93.9	75.5	63.4	88.6	75.3	61.8	83.7	75.0	34.3	86.9	33.1	84.3	32.1	82.7	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.7	23.0	20.7	-14.2	100.0	6.6	3.8	-18.9	102.7	-22.8	105.0	-26.5	107.1	-31.2	109.9		
			-13.1	72.3	7.1	2.3	-18.2	73.9	-22.3	75.2	-26.3	76.5	-31.4	78.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	47.0	17.7	18.3	30.0	49.1	62.5	71.3	76.0	73.6	61.6	46.4	32.6	22.5	(6)
(7)		DBStd	23.10	11.40	12.10	9.39	9.05	8.83	7.44	6.95	8.27	8.16	8.52	9.62	10.77	(7)
(8)		HDD50	4200	1000	889	621	123	11	0	0	8	172	522	854		(8)
(9)		HDD65	7568	1465	1309	1085	480	156	23	4	20	158	578	971	1319	(9)
(10)		CDD50	3089	0	0	1	97	397	640	805	731	357	60	1	0	(10)
(11)		CDD65	982	0	0	0	3	77	213	344	286	58	1	0	0	(11)
(12)		CDH74	9323	0	0	0	21	611	1937	3540	2830	379	5	0	0	(12)
(13)		CDH80	3772	0	0	0	1	174	753	1543	1210	91	0	0	0	(13)
(14)	Wind	WSAvg	10.4	10.8	11.5	11.5	10.8	10.3	10.0	9.3	9.5	9.6	10.3	10.6	11.1	(14)
(15)	Precipitation	PrecAvg	14.9	1.2	0.9	0.9	0.9	1.3	1.7	1.4	1.3	1.4	0.9	1.4	1.4	(15)
(16)		PrecMax	23.4	2.7	3.6	3.0	2.5	3.7	4.0	4.8	5.2	3.5	2.9	3.1	3.3	(16)
(17)		PrecMin	7.5	0.1	0.1	0.0	0.1	0.0	0.5	0.2	0.0	0.0	0.0	0.2	0.5	(17)
(18)		PrecStd	3.7	0.6	0.7	0.6	0.6	1.1	1.0	1.1	1.1	1.1	0.7	0.8	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.2	39.5	56.9	77.0	89.1	98.1	99.6	98.9	88.0	72.2	55.4	43.0	(19)
(20)		MCWB	38.6	36.2	46.0	56.9	62.0	67.2	68.1	67.7	62.2	58.7	50.7	41.6		(20)
(21)		2%	DB	36.3	36.4	49.6	72.4	84.5	92.0	94.9	94.3	82.2	66.5	51.1	40.4	(21)
(22)		MCWB	35.0	34.4	42.4	56.1	61.9	66.2	67.6	66.3	61.1	54.3	47.7	39.1		(22)
(23)		5%	DB	34.2	34.7	44.6	67.2	80.5	87.6	91.6	90.8	78.4	62.0	47.9	37.3	(23)
(24)		MCWB	32.9	33.1	39.3	52.8	60.9	65.6	67.0	65.9	59.9	52.6	45.1	36.1		(24)
(25)		10%	DB	32.7	33.2	41.1	62.9	76.5	84.0	88.3	86.7	74.5	58.0	44.4	34.8	(25)
(26)		MCWB	31.7	31.9	37.1	50.7	59.8	64.4	66.1	65.0	58.3	51.0	42.1	33.6		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.8	36.9	47.4	60.0	66.6	71.1	71.9	72.1	65.8	61.0	51.8	42.1	(27)
(28)		MCDWB	39.6	38.8	54.3	71.3	81.3	88.1	89.6	86.8	77.8	68.7	54.3	42.6		(28)
(29)		2%	WB	35.4	34.8	42.9	57.0	64.2	69.0	70.2	69.5	63.7	57.1	48.5	39.1	(29)
(30)		MCDWB	36.0	36.0	48.2	68.2	80.3	84.1	87.3	85.7	75.2	63.2	50.8	40.2		(30)
(31)		5%	WB	33.2	33.4	39.9	54.5	62.4	67.4	69.1	68.0	61.8	54.1	45.3	36.2	(31)
(32)		MCDWB	34.0	34.4	44.0	65.4	76.9	82.6	85.4	83.8	74.4	59.8	47.5	37.0		(32)
(33)		10%	WB	31.8	32.2	37.3	52.0	60.7	66.0	67.9	66.6	59.9	51.6	42.4	33.8	(33)
(34)		MCDWB	32.6	33.1	40.5	61.2	74.1	80.7	83.3	82.8	71.9	57.1	44.2	34.6		(34)
(35)	Mean Daily Temperature Range	MDBR	8.7	9.7	10.5	15.0	16.9	16.8	17.2	17.1	15.6	11.9	8.6	8.0		(35)
(36)		5% DB	MCDBR	6.6	8.2	15.3	19.7	20.9	20.4	20.1	19.7	19.9	17.2	11.3	6.5	(36)
(37)		MCWBR	6.2	7.4	10.0	9.2	8.5	7.4	6.4	6.1	8.1	9.4	8.7	6.4		(37)
(38)		5% WB	MCDBR	6.3	7.8	14.1	18.0	19.1	18.1	18.0	17.7	17.0	15.1	10.5	6.3	(38)
(39)		MCWBR	6.3	7.5	9.9	9.4	9.1	7.6	6.8	7.1	8.7	9.8	9.1	6.4		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.314	0.345	0.376	0.440	0.409	0.396	0.411	0.426	0.382	0.355	0.332	0.314		(40)
(41)		taud	2.111	2.026	2.114	2.082	2.270	2.348	2.307	2.242	2.341	2.360	2.291	2.152		(41)
(42)		Ebn at Noon	217	240	258	254	269	274	267	256	256	241	214	198		(42)
(43)		Edh at Noon	26	37	40	47	41	38	39	40	32	26	22	22		(43)
(44)	All-Sky Solar Radiation	RadAvg	295	623	1015	1420	1876	2031	1985	1734	1221	674	299	191		(44)
(45)		RadStd	58	137	159	116	169	183	139	128	116	95	52	42		(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+277	+140

Nomenclature: See separate page